

Abstract

1. The distance matrix $\mathcal{D}$ has a graph form: there is a weighted graph G_D on the m leaves whose effective resistances are the entries of $\mathcal{D}$, and its Laplacian L_D satisfies $\mathcal{B} := H\mathcal{D}H = -2L_D^+$.
2. The vector the algorithm reads, the largest-magnitude eigenvector $u^{(1)}$ of $\mathcal{B}$, is the Fiedler vector of L_D , so its signs cut the leaves along a single edge of the tree.
3. Stating a recovery guarantee needs two numbers of that vector and that spectrum: the margin $\delta_* = \min_i |u_i^{(1)}|$ and the gap $\Delta\lambda = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$.
4. Observing each distance with probability p and bounding the entrywise eigenvector error recovers the split exactly at $p = \Theta\left(\frac{d_{\max}^2}{\Delta\lambda^2\delta_*^2} \log m\right)$.
 - 4.1. This assumes $u^{(1)}$ is delocalized: $\delta_* \geq c_\delta/\sqrt{m}$ and $\|u^{(1)}\|_\infty \leq C_\delta/\sqrt{m}$.
 - 4.2. Putting $\Delta\lambda \asymp m$ and $\delta_* \asymp 1/\sqrt{m}$ into that rate gives $p \gtrsim \log m/m$.
5. What is open is how to read $\Delta\lambda$ and δ_* off a tree.
 - 5.1. One way in is to relate $\mathcal{D}$ to the similarity matrix $S = e^{-\mathcal{D}}$, so that what is known about L_S carries over.
 - 5.2. The other is to measure both numbers on simulated trees.

1 The Distance Matrix and Its Leading Vector

Two results from the literature carry the argument: the Schur complement results of Stone and Griffing [?], which say what the vector looks like on a tree, and the leave-one-out bound of Abbe et al. [?], which says how far sampling can move it.

Let T be a weighted tree with m leaves, and let $\mathcal{D}$ be the additive path-distance matrix between the leaves. We apply double centering with the orthogonal projector $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$ to form the distance operator:

$$\mathcal{B} := H\mathcal{D}H \quad (1)$$

The recovery algorithm classifies leaves by the sign pattern of $u^{(1)}$, the unit eigenvector of $\mathcal{B}$ belonging to its largest-magnitude eigenvalue.

Notation. Two weighted graphs on the m leaves appear throughout, and neither of their Laplacians is $\mathcal{B}$:

- G_S , with edge weights $S_{ij} = e^{-\mathcal{D}_{ij}}$, and its Laplacian $L_S = \text{Deg}(S) - S$; this is the operator the similarity route uses.
- G_D , the graph on the leaves whose effective resistances are the distances $\mathcal{D}_{ij}$, and its Laplacian L_D ; ?? builds it from the tree and shows $\mathcal{B} = -2L_D^+$, so $\mathcal{B}$ is the pseudoinverse of L_D up to a factor, not L_D itself.

L_T denotes the Laplacian of the tree T itself, on all of its vertices, and $\text{Deg}(W) = \text{diag}(W\mathbf{1})$ for any W .

Give each branch e of T its length $\ell_e > 0$ and read T as an electrical network in which branch e carries conductance $1/\ell_e$. Let L_T be the Laplacian of that weighted tree, let $R \subset V$ be its internal vertices, and let

$$L_D := L_T / (L_T)_{RR} \quad (2)$$

be the Schur complement of L_T onto the m leaves, taken with respect to its block $(L_T)_{RR}$ on the internal vertices. With those conductances the effective resistance between two vertices of a tree is the sum of the branch lengths along the path joining them [?], which is exactly $\mathcal{D}_{ij}$.

Proposition 1 (Identification).

- (i) L_D is the Laplacian of a connected weighted graph G_D on the leaves, and it carries the same effective resistances as T : $\mathcal{D}_{ij} = (L_D^+)_{ii} + (L_D^+)_{jj} - 2(L_D^+)_{ij}$.
- (ii) $\mathcal{B} = H\mathcal{D}H = -2L_D^+$. In particular $\mathcal{B}$ is negative semi-definite, which is why its largest-magnitude eigenvalue is its most negative one.
- (iii) If the smallest positive eigenvalue of L_D is simple — equivalently, if the gap $\Delta\lambda$ of ?? is positive — then $u^{(1)}$ is the Fiedler vector of G_D , up to sign.

Proof. (i) The Schur complement of a Laplacian with respect to a set of vertices is again a Laplacian, of the Kron-reduced graph, and Kron reduction leaves the effective resistances between the retained vertices unchanged [?]; connectedness passes from T to G_D . The stated identity is then the standard expression of effective resistance through the Laplacian pseudoinverse [?], applied to G_D .

(ii) Write $\pi = \text{diag}(L_D^+)$, so that (i) reads $\mathcal{D} = \pi\mathbf{1}^\top + \mathbf{1}\pi^\top - 2L_D^+$. Since $H\mathbf{1} = 0$, both rank-one terms drop out under double centering and $H\mathcal{D}H = -2HL_D^+H$. The graph G_D is connected, so $L_D^+\mathbf{1} = 0$ and $HL_D^+H = L_D^+$. Hence $\mathcal{B} = -2L_D^+$, and $L_D^+ \succeq 0$ gives $\mathcal{B} \preceq 0$.

(iii) L_D and L_D^+ share their eigenvectors, an eigenvalue $\lambda > 0$ of L_D becoming $1/\lambda$ of L_D^+ and the kernel staying the kernel. So the eigenvector of L_D for its smallest positive eigenvalue is the eigenvector of L_D^+ for its largest, and by (ii) the eigenvector of $\mathcal{B}$ for its most negative eigenvalue, which is the largest in magnitude. That is $u^{(1)}$, and simplicity fixes it up to sign. $\square$

This correspondence between numerical taxonomy and phylogenetics is set out in [?]. With it, statements about Fiedler vectors of trees are statements about $u^{(1)}$.

Lemma 1 (Target Geometry, [?, Theorem 3.3]). *The signs of $u^{(1)}$ cut the leaves of T along a single edge: there is one edge e of T whose removal puts $\{i : u_i^{(1)} > 0\}$ and $\{i : u_i^{(1)} \leq 0\}$ in the two different components.*

Proof. Theorem 3.3 of [?] is stated for a connected weighted graph, for the Schur complement of its Laplacian with respect to the points of articulation, and for a Fiedler vector Y of that Schur complement. Here the graph is T , its points of articulation are the internal vertices R , the Schur complement is L_D of (??), and $Y = u^{(1)}$ by ??(iii). The theorem then supplies a subset $S \subseteq R$ such that $V_+ \cup S$ and $V_- \cup (R - S)$ both induce connected subgraphs of T , where $V_+ = \{i : u_i^{(1)} > 0\}$ and V_- is its complement among the leaves. Two complementary vertex sets of a tree that are each connected are joined by exactly one edge, and that edge is e . $\square$

The theorem puts a leaf with $u_i^{(1)} = 0$ on the non-positive side, so it gives the split without claiming that any entry is nonzero. The margin $\delta_* := \min_i |u_i^{(1)}|$ is a separate quantity, which we assume. For a tree that is not symmetric $u^{(1)}$ is not piecewise constant, and how far it is from constant enters only through δ_* .

Assumption 1 (Delocalization). *There are constants $c_\delta, C_\delta > 0$ with $\delta_* \geq c_\delta/\sqrt{m}$ and $\|u^{(1)}\|_\infty \leq C_\delta/\sqrt{m}$.*

The lower half of ?? is the one ?? leaves open, and everything below uses it wherever δ_* appears; ?? in Appendix ?? reports what it costs on each kind of tree.

Reading ?? eigenvalue by eigenvalue says what those two numbers are.

Corollary 1 (The two numbers are Fiedler data of G_D). *Let $0 < \mu_1 \leq \mu_2 \leq \dots \leq \mu_{m-1}$ be the positive eigenvalues of L_D and let y be the Fiedler vector of G_D . Then*

$$|\lambda_1(\mathcal{B})| = \frac{2}{\mu_1}, \quad |\lambda_2(\mathcal{B})| = \frac{2}{\mu_2}, \quad \Delta\lambda = 2 \frac{\mu_2 - \mu_1}{\mu_1 \mu_2}, \quad \frac{\Delta\lambda}{|\lambda_1(\mathcal{B})|} = 1 - \frac{\mu_1}{\mu_2}, \quad (3)$$

and $\delta_* = \min_i |y_i|$.

Proof. $\mathcal{B} = -2L_D^+$ by ??(ii), and L_D^+ has eigenvalues $1/\mu_k$ on the orthogonal complement of $\mathbf{1}$, so the k -th largest magnitude eigenvalue of $\mathcal{B}$ is $2/\mu_k$. The rest is arithmetic, and the last claim is ??(iii). $\square$

So the rate depends on three properties of one weighted graph on the leaves: its algebraic connectivity μ_1 , the ratio μ_1/μ_2 to the next eigenvalue, and the smallest entry of its Fiedler vector. A relative gap of 0.60 measured on Kingman trees says $\mu_1/\mu_2 = 0.40$ there.

For a single point of articulation r , Stone and Griffing bound the algebraic connectivity of the reduced graph between the two largest Perron values of the branches hanging at r : $\nu_1 < \mu^{-1} < \nu_0$ [?, Corollary 2.1], where $\nu_i = \rho(L_{C_i}^{-1})$ is taken over the connected components C_i at r . Through (??) that reads $2\nu_1 < |\lambda_1(\mathcal{B})| < 2\nu_0$, a two-sided bound on the leading eigenvalue in terms of the branches themselves. Extending it to the full internal set R , and adding the companion statement for μ_1/μ_2 and for $\min_i |y_i|$, is exactly the open question of ??.

2 Exact Recovery from Sampled Distances

2.1 Observation Model and Error Decomposition

We observe each pairwise distance independently with probability p . The inverse-probability weighted empirical matrix is $\hat{\mathcal{D}} = p^{-1}\mathcal{D} \circ \Omega$, where $\Omega_{ij} \sim \text{Bernoulli}(p)$. The empirical distance operator is $\hat{\mathcal{B}} = H\hat{\mathcal{D}}H$, with leading eigenvector $\hat{u}^{(1)}$.

The error takes the form $\hat{\mathcal{B}} = \mathcal{B} + E_{\mathcal{B}}$, where $E_{\mathcal{B}} = H(\hat{\mathcal{D}} - \mathcal{D})H$ is the zero-mean sampling noise. Let $E_{\mathcal{D}} = \hat{\mathcal{D}} - \mathcal{D}$ be the uncentered noise matrix, which has independent mean-zero entries bounded by $d_{\max}/p$, and variance bounded by $d_{\max}^2/p$.

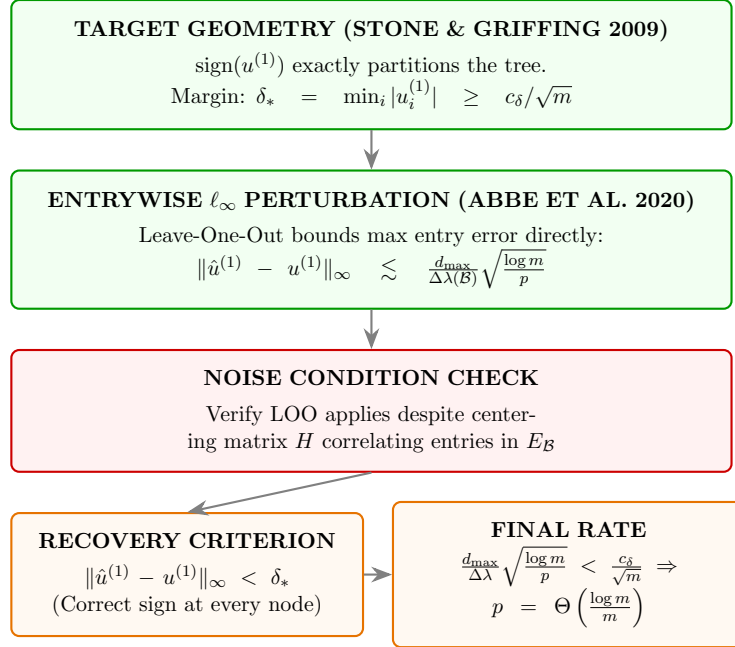


Figure 1: The Leave-One-Out (LOO) proof route for single-stage exact recovery.

2.2 Full Entrywise Proof

Theorem 1 (Single-Stage Exact Recovery). *Assume the spectral gap $\Delta\lambda = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$ is strictly positive. If the sampling rate satisfies*

$$p = \Theta\left(\frac{d_{\max}^2}{\Delta\lambda^2 \cdot \delta_*^2} \log m\right) \quad (4)$$

then with high probability $\text{sign}(\hat{u}^{(1)}) = \text{sign}(u^{(1)})$. For a tree with $\Delta\lambda \asymp m$ and $\delta_ \asymp 1/\sqrt{m}$ this is $p = \Theta(\frac{\log m}{m})$.*

Proof. We bound the largest entry of the eigenvector error. The tool is the leave-one-out (LOO) bound, and the part of it we use is [?]:

Theorem 2 (Abbe et al. 2020, Theorem 2.1 (Consequence for Symmetric Matrices)). *Let N be a symmetric matrix with eigendecomposition $N = U\Lambda U^\top$, and let $\hat{N} = N + E$ where E is a symmetric noise matrix with independent mean-zero entries (up to symmetry). Let $\hat{U}$ denote the empirical*

top r eigenvectors. Provided the spectral gap $\Delta\lambda$ is sufficiently large, there exists an orthogonal alignment matrix W such that:

$$\|\hat{U}W - U\|_\infty \lesssim \frac{\|EU\|_\infty}{\Delta\lambda} + \frac{\|E\|_2\|E\|_{\text{row_max}}}{\Delta\lambda^2}\|U\|_\infty \quad (5)$$

Here we use it with $r = 1$: U is the single vector $u^{(1)}$, W is a sign flip, which we fix by taking the signs to be aligned, N is $\mathcal{B}$, and E is the centered noise $E_{\mathcal{B}}$. The matrix $\mathcal{B}$ is full rank, but only the gap $\Delta\lambda$ at the top of its spectrum enters the LOO expansion, which gives

$$\|\hat{u}^{(1)} - u^{(1)}\|_\infty \lesssim \frac{\|E_{\mathcal{B}}u^{(1)}\|_\infty}{\Delta\lambda} + \frac{\|E_{\mathcal{B}}\|_2\|E_{\mathcal{B}}\|_{\text{row_max}}}{\Delta\lambda^2}\|u^{(1)}\|_\infty \quad (6)$$

We now bound each norm on the right for the centered noise matrix $E_{\mathcal{B}} = HE_{\mathcal{D}}H$.

1. Bounding the first-order term $\|E_{\mathcal{B}}u^{(1)}\|_\infty$:

Since the population vector is centered ($Hu^{(1)} = u^{(1)}$), we have $E_{\mathcal{B}}u^{(1)} = HE_{\mathcal{D}}u^{(1)}$. The i -th entry of $E_{\mathcal{D}}u^{(1)}$ is:

$$(E_{\mathcal{D}}u^{(1)})_i = \sum_{j=1}^m (E_{\mathcal{D}})_{ij}u_j^{(1)} \quad (7)$$

Because the entries $(E_{\mathcal{D}})_{ij}$ are independent with mean zero, and $u^{(1)}$ is a fixed vector, this is a sum of independent random variables. The variance of the sum is bounded by $\sum_j \frac{d_{\max}^2}{p} (u_j^{(1)})^2 = \frac{d_{\max}^2}{p}$. The maximum absolute value of a single summand is bounded by $\frac{d_{\max}}{p}\|u^{(1)}\|_\infty$. By Bernstein's inequality, with high probability:

$$\|E_{\mathcal{D}}u^{(1)}\|_\infty \lesssim d_{\max}\sqrt{\frac{\log m}{p}} \quad (8)$$

The centering matrix H only subtracts the mean $\frac{1}{m}\mathbf{1}^\top E_{\mathcal{D}}u^{(1)}$, which is an average of these concentrated sums and is therefore of even smaller order. Thus, $\|E_{\mathcal{B}}u^{(1)}\|_\infty \lesssim d_{\max}\sqrt{\frac{\log m}{p}}$.

2. Bounding the operator and row norms ($\|E_{\mathcal{B}}\|_2$ and $\|E_{\mathcal{B}}\|_{\text{row_max}}$):

By standard Matrix Bernstein inequalities [?], the operator norm of the independent noise matrix concentrates as $\|E_{\mathcal{D}}\|_2 \lesssim d_{\max}\sqrt{\frac{m}{p}}$. Because H is a projector, $\|E_{\mathcal{B}}\|_2 \leq \|E_{\mathcal{D}}\|_2 \lesssim d_{\max}\sqrt{\frac{m}{p}}$. Similarly, the row norm $\|e_i^\top E_{\mathcal{B}}\|_2$ is dominated by the Euclidean norm of a row of $E_{\mathcal{D}}$, which is the square root of m independent variables each with variance $d_{\max}^2/p$. By scalar concentration, $\|E_{\mathcal{B}}\|_{\text{row_max}} \lesssim d_{\max}\sqrt{\frac{m}{p}}$.

3. Synthesis and Recovery Criterion:

Putting the three bounds of steps 1 and 2 into the LOO bound leaves two terms:

$$\|\hat{u}^{(1)} - u^{(1)}\|_\infty \lesssim \underbrace{\frac{d_{\max}}{\Delta\lambda}\sqrt{\frac{\log m}{p}}}_{\text{(I) linear in the noise}} + \underbrace{\frac{d_{\max}^2 m}{p \Delta\lambda^2}\|u^{(1)}\|_\infty}_{\text{(II) quadratic in the noise}} \quad (9)$$

The two are of different kinds. Term (I) is the noise acting once, along the population vector: it comes from $\|E_{\mathcal{B}}u^{(1)}\|_\infty$, one Bernstein sum per coordinate. Term (II) is the noise acting twice, through the product $\|E_{\mathcal{B}}\|_2\|E_{\mathcal{B}}\|_{\text{row_max}}$, which is why it carries $\Delta\lambda^{-2}$ and a second factor of $d_{\max}$.

A sign is read correctly at every leaf as soon as the total deviation stays below the margin, so we ask each term to be smaller than $\delta_*/2$. With the delocalization of ??, $\|u^{(1)}\|_\infty \asymp 1/\sqrt{m}$ and $\delta_* \asymp 1/\sqrt{m}$, this asks

$$(I) : \quad \frac{d_{\max}}{\Delta\lambda} \sqrt{\frac{\log m}{p}} < \frac{c_\delta}{2\sqrt{m}} \quad \implies \quad p \gtrsim \frac{d_{\max}^2 m \log m}{\Delta\lambda^2}, \quad (10)$$

$$(II) : \quad \frac{d_{\max}^2 m}{p \Delta\lambda^2 \sqrt{m}} < \frac{c_\delta}{2\sqrt{m}} \quad \implies \quad p \gtrsim \frac{d_{\max}^2 m}{\Delta\lambda^2}. \quad (11)$$

The two requirements differ by exactly the factor $\log m$, so (??) is the binding one and (??) is satisfied automatically at that rate. Hence

$$p = \Theta\left(\frac{d_{\max}^2 m \log m}{\Delta\lambda^2}\right) \quad (12)$$

is enough for $\text{sign}(\hat{u}^{(1)}) = \text{sign}(u^{(1)})$ at every leaf, and with $\Delta\lambda \asymp m$ this is the $p = \Theta(\log m/m)$ of the statement. $\square$

2.3 Open Question: How Do $\Delta\lambda$ and δ_* Depend on the Tree?

The rate above is written in terms of two numbers, and both can be derived once the tree is fixed:

- the gap $\Delta\lambda = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$, which says how far the leading direction of $\mathcal{B}$ stands out from the next one;
- the smallest entry $\delta_* = \min_i |u_i^{(1)}|$, which says how close the closest leaf comes to sitting on the fence between the two sides.

We put $\Delta\lambda \asymp m$ and $\delta_* \asymp 1/\sqrt{m}$ into the general rate to get $p \gtrsim \log m/m$. What is open is how to read those two numbers off a tree:

Given a weighted tree T with m leaves, how small can $\Delta\lambda$ and δ_* be, in terms of the shape of T and the lengths of its branches?

?? already puts the question in a more familiar form. Since $\Delta\lambda/|\lambda_1| = 1 - \mu_1/\mu_2$ and $\delta_* = \min_i |y_i|$, what is wanted is a lower bound on the algebraic connectivity μ_1 of the reduced graph G_D , a bound on the ratio μ_1/μ_2 , and a lower bound on the smallest entry of its Fiedler vector — three questions about one weighted graph built from the tree. ?? is the other way in: it reaches the same two numbers from the similarity Laplacian, at the price of the row-sum spread and a term of the order of the squared distances. What each kind of tree does to them is collected in Appendix ??.

3 The Connection Between S and $\mathcal{D}$

The other way to cut a tree from the same data is to turn distances into similarities and take the Fiedler vector of the similarity Laplacian L_S [?].

Write $S = e^{-\mathcal{D}}$ entrywise, with $S_{ii} = 1$, and $L_S = \text{Deg}(S) - S$.

By Taylor expansion of e^{-t} around $t = 0$,

$$S = J - \mathcal{D} + \Gamma, \quad \Gamma_{ij} := e^{-\mathcal{D}_{ij}} - 1 + \mathcal{D}_{ij}, \quad 0 \leq \Gamma_{ij} \leq \frac{1}{2} \mathcal{D}_{ij}^2, \quad (13)$$

with J the all-ones matrix. Applying $W \mapsto \text{Deg}(W) - W$ term by term,

$$L_S = (mI - J) - (\text{Deg}(\mathcal{D}) - \mathcal{D}) + (\text{Deg}(\Gamma) - \Gamma). \quad (14)$$

Centering brings in the distance operator. With $\text{Deg}(\mathcal{D}) = \text{diag}(d)$, $d_i = \sum_j \mathcal{D}_{ij}$, and $\bar{d}$ the mean of the d_i , and using $HJH = 0$ and $H\mathcal{D}H = \mathcal{B}$,

$$HL_S H = (m - \bar{d})H + \mathcal{B} - H\text{diag}(d - \bar{d})H + H(\text{Deg}(\Gamma) - \Gamma)H, \quad (15)$$

with no approximation so far.

Lemma 2 (When the two routes share an eigenvector). *If $d_i = \bar{d}$ for every leaf and $\Gamma = 0$, then $HL_S H = (m - \bar{d})H + \mathcal{B}$, and the Fiedler vector of L_S is $u^{(1)}$.*

Proof. The last two terms of (??) vanish, so on the centered subspace the two operators differ by a multiple of the identity. Their eigenvectors coincide, and the smallest eigenvalue of L_S there belongs to the most negative eigenvalue of $\mathcal{B}$. $\square$

So the two routes are separated by the row-sum spread $\max_i |d_i - \bar{d}|$ and by Γ , which is second order in the distances; Appendix ?? reports how far apart the two vectors sit in practice.

References

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A Answering the Open Question by Simulation

Jukes–Cantor sequences of length $L = 10^4$ on trees with $m = 1000$ leaves, in two pools: Kingman coalescent and pure birth (rate 0.5, no extinction). Split statistics use 1000 trees per model, spectra and vectors 200, the route comparison 15. Entries are medians, with the quartile range in parentheses.

Model	$\mathcal{B} = HDH$			L_S	
	$ \lambda_1 $	$ \lambda_2 $	$(\lambda_1 - \lambda_2)/ \lambda_1 $	λ_2	$(\lambda_3 - \lambda_2)/\lambda_3$
Kingman	352.1	114.9	0.60 (0.44, 0.80)	244.2	0.20 (0.08, 0.40)
birth–death	1200.8	721.1	0.38 (0.27, 0.47)	0.0260	0.25 (0.11, 0.44)

Table 1: Leading spectrum of each operator, 200 trees per model. $\mathcal{B}$ is negative semi-definite, so its eigenvalues are ordered by magnitude; L_S is ordered ascending, and $\lambda_1(L_S) = 0$.

Model	$\sqrt{m} \delta_*$	$\sqrt{m} \ u^{(1)}\ _\infty$	$\sqrt{m} \ v^{(2)}\ _\infty$
Kingman	0.356 (0.068, 0.659)	1.59	1.9
birth–death	0.0241 (0.00038, 0.308)	1.80	2.0

Table 2: Entries of the classifying vectors, 200 trees per model, each vector normalised to unit length and scaled by $\sqrt{m}$; the range in parentheses is the 10th to 90th percentile. $\delta_* = \min_i |u_i^{(1)}|$ and $v^{(2)}$ is the Fiedler vector of L_S .

Model	valid edge (%)		median η		same split (%)	$ \langle u^{(1)}, v^{(2)} \rangle $	$\max_i d_i - \bar{d} /\bar{d}$
	$\mathcal{B}$	L_S	$\mathcal{B}$	L_S			
Kingman	98.6	99.6	1.85	3.00	64	0.98	0.49
birth–death	55.5	99.6	1.53	3.08	33	0.97	0.17

Table 3: Which split each operator reads, 1000 trees per model; the last two columns are over 15 trees per model. A split is a valid edge when it is a single-edge bipartition of the true tree, and η is the ratio of the two clan sizes.

Each panel of the figures below carries its own scale: the four matrices are on separate colour scales, the eigenvalue histograms clip the $\lambda_1(L_S) \equiv 0$ bar, and the entry histograms use logarithmic counts with dotted lines at ± 1 .

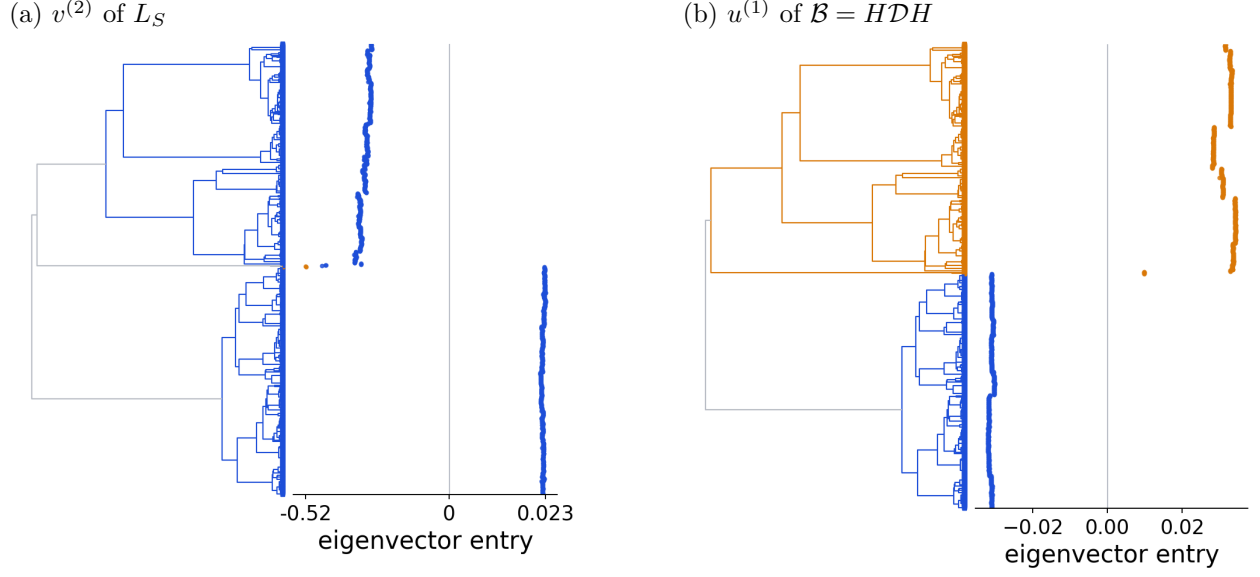


Figure 2: One Kingman tree, cut by each vector; the pool's maximum-disagreement tree, split 505/495 by $u^{(1)}$ and 998/2 by $v^{(2)}$.

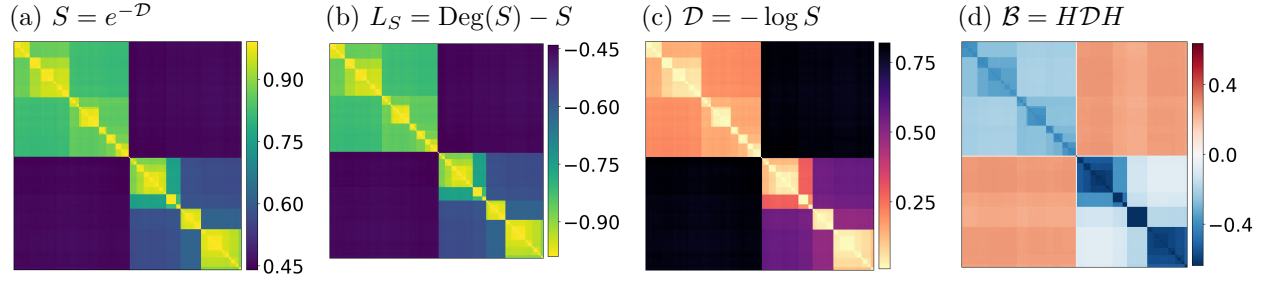


Figure 3: The four matrices on the tree of ??, leaves in clade order; the degree diagonal of L_S is masked.

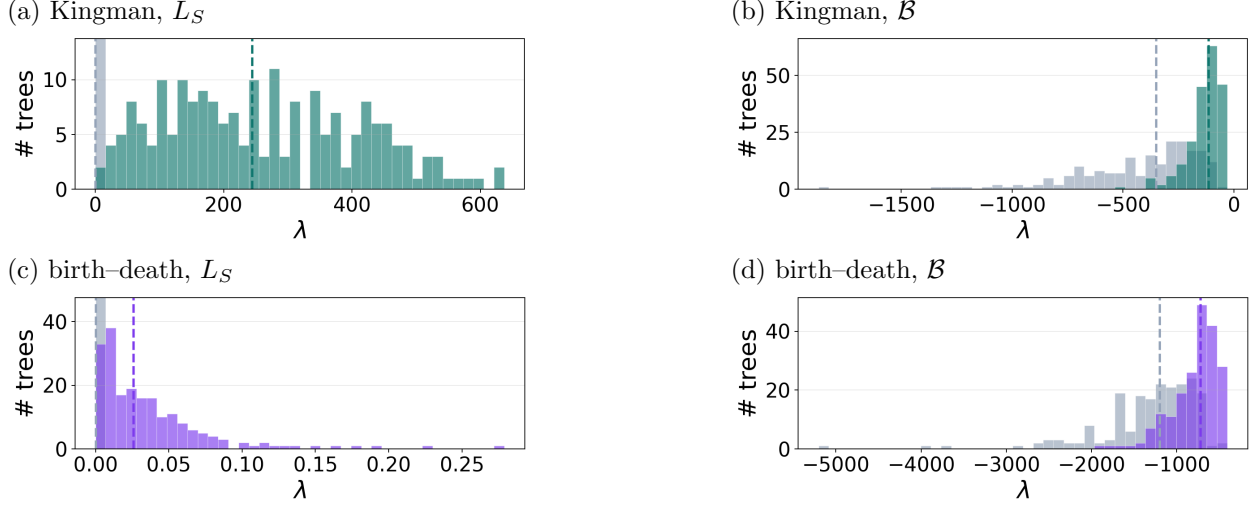


Figure 4: First two eigenvalues per model and route, 200 trees each: grey λ_1 , coloured λ_2 , dashed at the medians (??).

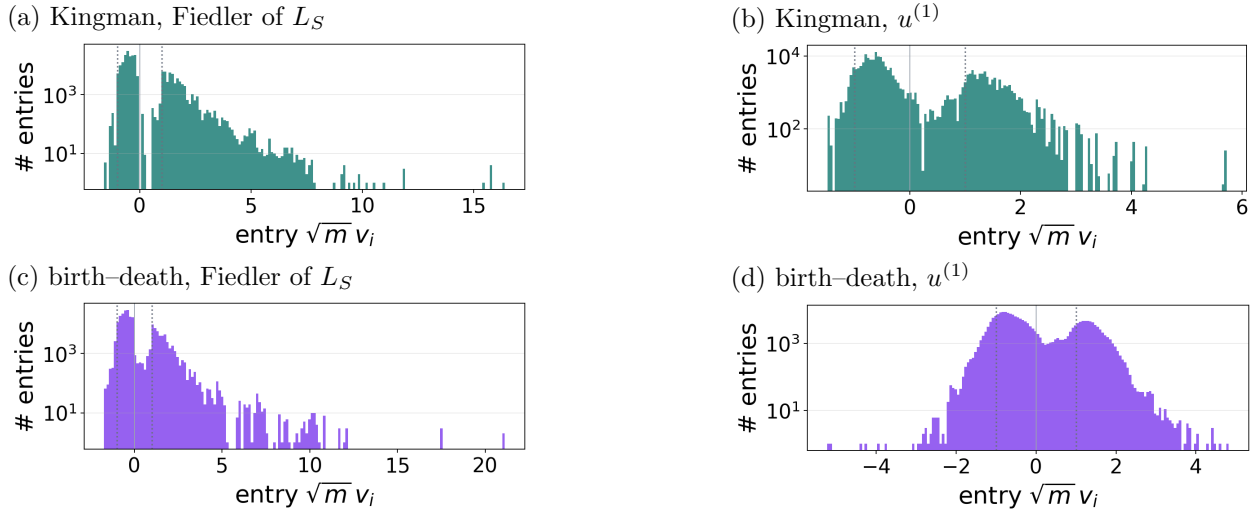


Figure 5: Pooled entries of the classifying vectors, $\sqrt{m}$ -scaled and oriented minority-positive; pool maxima 5.7 for $u^{(1)}$ and 16.4 (21.1) for $v^{(2)}$ on Kingman (birth-death) trees, against $\sqrt{m} = 31.6$ (??).

Bottom line. The upper half of ?? holds on both models ($\sqrt{m}\|u^{(1)}\|_\infty \approx 1.6$ and 1.8), the lower half holds on Kingman trees with a small constant ($\sqrt{m}\delta_* = 0.36$) and fails on pure-birth trees (0.024, with a tenth of them below 0.0004).

B Answering the Open Question by Theory

Tree	$\delta_* \approx$	$\Delta\lambda \approx$	Note
Balanced binary	$1/\sqrt{m}$	order m	Symmetry fixes the vector: $u^{(1)}$ is two-valued with $\alpha = 1/2$, so every entry has the same size and the margin is as large as it can be.
Kingman coalescent	$\sqrt{\alpha/(1-\alpha)}/\sqrt{m}$	a constant fraction of $ \lambda_1 $	Theory. The topology is Yule, whose root split is uniform in size $[\cdot, \cdot]$, so the smaller side is $\alpha \sim \text{U}\{1/m, \dots, 1/2\}$: typically of order one, with a tail down to $1/m$ when the split peels off a single leaf. The last coalescence lasts $\text{Exp}(1)$ against a tree height of mean $2(1 - 1/m)$ $[\cdot, \cdot]$, so the deepest split stands clear of the next one by a fixed share of $ \lambda_1 $ rather than a vanishing one. Measured values in ????
Pure birth (Yule branch lengths)	$0.024/\sqrt{m}$	$0.38 \lambda_1 $	Measured, 200 trees at $m = 1000$ (????). The topology law is the same as the row above, so α is drawn from the same distribution; what changes is where the branch lengths sit, and that alone moves the median margin down by a factor of 15 against the coalescent's $0.36/\sqrt{m}$.
Caterpillar, very uneven branch lengths (extreme)	well below $1/\sqrt{m}$	well below m	The leading vector puts most of its weight on a handful of leaves, which is $\alpha \rightarrow 1/m$ in (??), and the remaining entries are what sets δ_* . How far below $1/\sqrt{m}$ the margin goes, and how far below m the gap goes, is the open part of the question.

Table 4: What each kind of tree does to the two numbers the rate is written in.

Reference point for the δ_* column: a two-valued $u^{(1)}$ that puts a fraction $\alpha \leq 1/2$ of the leaves on the smaller side has

$$\delta_* = \frac{1}{\sqrt{m}} \sqrt{\frac{\alpha}{1-\alpha}}. \quad (16)$$